

AI ROUND – SOFTWARE ENGINEER (FRESHER)

Trading Dashboard – Practical Coding Assignment

Scenario

You are joining a fintech team developing a simplified **Trading Dashboard** for retail investors.

As part of this AI coding round, you need to build the same core experience for:

- **Web:** Next.js
- **Mobile:** React Native

You may use AI tools such as ChatGPT, GitHub Copilot, Claude, Amazon Q, or any other coding assistant.

1. Web Application – Next.js

Build a responsive Trading Dashboard using **Next.js + TypeScript**.

The dashboard should include:

Market Overview

Display:

- NIFTY 50
- SENSEX
- One additional index

Show current value, change, percentage change, and market status.

Watchlist

Create a watchlist containing at least **6 stocks**.

Display:

- Stock symbol
- Company name
- Current price
- Price change
- Percentage change

Live / Simulated Prices

Use any of the following:

- Free/public API

- Dummy API
- Mock JSON data
- Locally simulated price updates

Prices should update periodically to simulate live market data.

Search & Stock Details

Allow users to:

- Search stocks by name or symbol
- Select a stock
- View price and percentage change
- View a simple intraday price chart

Application States

Handle:

- Loading
 - API/data failure
 - No search results
 - Empty data
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2. Mobile Application – React Native

Build the same core Trading Dashboard experience using **React Native**.

The mobile application should include:

- Market overview
- Watchlist
- Stock search
- Stock details
- Current price
- Price change
- Percentage change
- Intraday chart
- Simulated/live price updates
- Loading and error states

The UI should follow **mobile-friendly design and interaction patterns**.

3. AI Usage

This is an **AI Coding Round**, so you are expected to use an AI coding assistant.

You should be able to explain:

- How you used AI during development
- Why you selected your technical approach
- How you verified AI-generated code
- Any issues found in AI-generated code
- How you debugged or improved the solution

Do not blindly copy and paste AI-generated code. You should understand and explain your implementation.

4. Final Submission

Submit:

- Working **Next.js application**
- Working **React Native application**
- Git repository/source code
- README containing:
 - Setup instructions
 - How to run the projects
 - API/data source
 - Key assumptions
 - Libraries used
 - Known limitations

Also mention which AI tools you used and how they helped during development.

5. Suggested Duration

Total: 90–120 minutes

The solution does not need to be production-ready.

Focus on:

Problem solving • Coding fundamentals • UI implementation • API/data handling • Debugging • Effective AI usage